(i)

$$\begin{aligned}\frac{1}{r+1}-\frac{1}{r}+\frac{1}{r^2} &= \frac{r^2-r(r+1)+(r+1)}{r^2(r+1)}\\ &= \frac{r^2-r^2-r+r+1}{r^2(r+1)}\\ &= \frac{1}{r^2(r+1)}\end{aligned}$$

$$\begin{aligned}\sum_{r=1}^{N}\frac{1}{r^2(r+1)} &= \sum_{r=1}^{N}\left(\frac{1}{r+1}-\frac{1}{r}\right)+\sum_{r=1}^{N}\frac{1}{r^2}\\ &= \left(\frac{1}{N+1}-1\right)+S_N\\ &= \sum_{r=1}^{N}\frac{1}{r^2}-1+\frac{1}{N+1}\end{aligned}$$

As $N\to\infty$, $\dfrac{1}{N+1}\to 0$ and $S_N\to\dfrac{1}{6}\pi^2$

$$\implies \boxed{\sum_{r=1}^{\infty}\frac{1}{r^2(r+1)}=\frac{1}{6}\pi^2-1}$$

(ii)

$$\frac{1}{r^2(r+1)}-\frac{1}{r(r+1)(r+2)}=\frac{(r+2)-r}{r^2(r+1)(r+2)}=\frac{2}{r^2(r+1)(r+2)}$$

$$\frac{1}{r(r+1)(r+2)}=\frac{1}{2}\left(\frac{1}{r(r+1)}-\frac{1}{(r+1)(r+2)}\right)$$

$$\implies \sum_{r=1}^{N}\frac{1}{r(r+1)(r+2)}=\frac{1}{2}\left(\frac{1}{2}-\frac{1}{(N+1)(N+2)}\right)$$

$$\begin{aligned}\sum_{r=1}^{N}\frac{1}{r^2(r+1)(r+2)} &= \frac{1}{2}\sum_{r=1}^{N}\frac{1}{r^2(r+1)}-\frac{1}{2}\sum_{r=1}^{N}\frac{1}{r(r+1)(r+2)}\\ &= \frac{1}{2}\left(S_N-1+\frac{1}{N+1}\right)-\frac{1}{4}\left(\frac{1}{2}-\frac{1}{(N+1)(N+2)}\right)\\ &= \frac{1}{2}S_N-\frac{5}{8}+\frac{1}{2(N+1)}+\frac{1}{4(N+1)(N+2)}\\ &= \frac{1}{2}S_N-\frac{5}{8}+\frac{2N+5}{4(N+1)(N+2)}\end{aligned}$$

As $N \to \infty$, $\dfrac{2N+5}{4(N+1)(N+2)} \to 0$

$$\implies \boxed{\sum_{r=1}^{\infty} \frac{1}{r^2(r+1)(r+2)} = \frac{1}{12}\pi^2 - \frac{5}{8}}$$

(iii)

$$\frac{1}{r^2(r+1)^2} = \left(\frac{1}{r} - \frac{1}{r+1}\right)^2 = \frac{1}{r^2} + \frac{1}{(r+1)^2} - \frac{2}{r(r+1)}$$

$$\because \frac{2}{r^2(r+1)} = \frac{2}{r^2} - \frac{2}{r(r+1)}$$

$$\therefore \frac{2}{r^2(r+1)} - \frac{1}{r^2(r+1)^2} = \frac{1}{r^2} - \frac{1}{(r+1)^2}$$

$$\sum_{r=1}^{N} \left(\frac{1}{r^2} - \frac{1}{(r+1)^2}\right) = 1 - \frac{1}{(N+1)^2} \implies 1$$

Both $\sum \dfrac{2}{r^2(r+1)}$ and $\sum \dfrac{1}{r^2(r+1)^2}$ converge, thus

$$\boxed{\sum_{r=1}^{\infty} \frac{1}{r^2(r+1)^2} = \sum_{r=1}^{\infty} \frac{2}{r^2(r+1)} - 1}$$

This solution was submitted by **Jon**
Published and refined by **Jon**
Last updated: September 2, 2026